

Transcendence of a planar reflection–group growth series and its relaxed companion

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Abstract

Let $W = \langle R_0, R_1, R_2 \rangle$ be the group generated by the reflections in the three sides of a right triangle in $\mathbb{Q}^2$ with positive unequal legs, and let $U(x) = \sum_{n \geq 0} u_n x^n$ be its geodesic word-length growth series — the integer sequence OEIS A396406 — with relaxed companion V . We prove:

- (i) the catalytic building-block q -series $\Sigma_0, \Sigma_1, S_0, S_1$ are transcendental over $\mathbb{Q}(q)$, *unconditionally*: their integer coefficients grow super-polynomially along the squares, so $|q| = 1$ is a natural boundary by the Pólya–Carlson theorem;
- (ii) the common growth rate is $\beta_2 = 1.4916177871 \dots$ *exactly*, and $\frac{3}{2}$ is excluded;
- (iii) the relaxed series V is transcendental over $\mathbb{Q}(x)$; its one analytic input, a simple-saddle steepest-descent estimate, we prove here from first principles (Cauchy’s theorem, Stirling, elementary Gaussian integration) rather than cite, with every constant explicit. For the true series U we prove: *conditional on a single amplitude estimate* ($\star$) *at the pole*, U is transcendental over $\mathbb{Q}(x)$, on the same analytic footing as V . By an exact machine-checked identity U reduces to V ’s estimate plus ($\star$); the amplitude has an elementary closed-form expansion with exact rational constants (Remark 7), and matching its zero against the travel-pole equation gives $\delta w = \frac{\sqrt{2}}{8} \tau^{3/2}$, hence $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the gate threshold $1/\sqrt{2}$ — which *establishes* ($\star$) *to leading order* and at the first 70 poles; its residual $O(\tau)$ term we verify numerically but do not prove. The turning-point amplitude atom on the U -side, long the special-function obstacle, is an explicit rational inequality *formally verified in Lean 4 / Mathlib* (Lemma 25);
- (iv) an orthogonal, analysis-free route via Christol’s theorem would settle U without any of the above: the p -kernel of $(u_n \bmod p)$ is machine-verified to be maximally non-automatic at $p = 3, 5$, though a proof for dense-support series is open.

Thus V rests on self-contained analysis, and U on the same analysis plus the single residual ($\star$): every coefficient is derived elementarily and machine-verified, and the remainders are at the standard simple-saddle level — the precise rigor standing is stated in the text. In sharp contrast with the rational growth of hyperbolic groups and the algebraic growth series of Parry’s wreath products, the geometric word metric places this virtually-lamplighter group outside every algebraic class.

1 Introduction

A right triangle $T \subset \mathbb{Q}^2$ with legs $a \neq b$ has three sides; the reflections across them generate a group $W = \langle R_0, R_1, R_2 \rangle \leq \text{Aff}(\mathbb{R}^2)$. Its *growth series* $U(x) = \sum_n u_n x^n$ counts group elements by

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minimal word length in $\{R_0, R_1, R_2\}$. Because the legs are unequal the dihedral angles are irrational multiples of π (Niven), so W is a generic triangle group [7] rather than a Coxeter group, and — unlike the rational Coxeter case governed by Steinberg’s formula [15] — its growth is not a priori rational. The sequence $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \dots$ is OEIS A396406; it coincides with the Fibonacci numbers F_{n+3} for $1 \leq n \leq 9$ and first deviates at $n = 10$. This paper settles the arithmetic nature of both series: the relaxed companion V is transcendental over $\mathbb{Q}(x)$ with every analytic input proved self-contained, and the true series U is transcendental over $\mathbb{Q}(x)$ conditional on a single explicit amplitude estimate ($\star$) at the pole — derived to leading order and verified at the first 70 poles, its $O(\tau)$ residual the sole remaining gap (§4; Remarks 6, 7, and 28).

Context. For word-growth series of groups, rationality is the norm in the classical families: hyperbolic groups have rational growth series with respect to every finite generating set (Cannon [3]), Coxeter systems are governed by Steinberg’s rational formula, and for wreath products — including lamplighter-type groups — Parry computed the growth series for natural generating sets and found them *algebraic* [14]. Provably transcendental growth series are rare; the classical example is Stoll’s [16]: the higher Heisenberg groups admit both rational and transcendental growth series, the transcendental one arising from a carefully constructed generating set. The group studied here is virtually the lamplighter $\mathbb{Z}\wr\mathbb{Z}$ [1], and its generating set is not constructed but imposed by Euclidean geometry — the three side-reflections of a right triangle. The results below show that for this natural geometric system the growth series is transcendental, by an explicit mechanism (a natural boundary for the building blocks; an accumulation of poles for the series themselves). This places the example in sharp contrast with the algebraicity of Parry’s wreath-product series: the geometric word metric moves a virtually-lamplighter group outside every algebraic class. The higher-dimensional picture is subtler than a clean dimensional dichotomy [2]: the abstract right-angled Coxeter envelope of an n -orthoscheme has rational growth (rate $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$), but the geometric group is an amenable proper quotient that matches this rational envelope only to a finite depth before deviating below it. The plane is the one dimension in which the deviation has been computed past its onset — there it is the transcendental series treated here — and whether every higher dimension is likewise eventually transcendental is an open problem.

The relaxed/true dichotomy. The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller, analytically tamer *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\geq 0}$ a cycle count. The relaxed counting function is the companion V ; one has $v_n \geq u_n$, equality for $n \leq 7$, first strict at $n = 8$. Both are governed by a linear q -difference (dilation $t \mapsto q^2t$) *catalytic functional equation* (§2) that is q -holonomic and not of polynomial-kernel type.

Method. The catalytic transfer is assembled from four integer q -series $\Sigma_0, \Sigma_1, S_0, S_1$ with j^2 -gapped coefficients. Their coefficients grow super-polynomially, so Pólya–Carlson yields unconditional transcendence over $\mathbb{Q}(q)$ (§3). Both V and U acquire poles at the *travel poles* q_m , the roots of $\Sigma_1^T(q) = 1$; transcendence follows from accumulation of these poles at $q = 1$ once a certain auxiliary integral T_2 is shown to be small (§4). That estimate is proved *self-contained* by a two-saddle steepest-descent argument — no asymptotic theorem is cited — and it is the *single* analytic input, shared by V (§5) and U (§6).

The U -side carries one extra ingredient. Writing the cocycle off-diagonal P_{12} in exact closed form as a Hahn–Exton q -Bessel function $J_{3/2}^{(3)}$, the transcendence gate reduces to one amplitude bound on that series at the travel poles. This bound is *derived* (Remark 7): the series collapses exactly to

sinh-products, its expansion has exact rational constants, the pole equation collapses to the *same* sinh-family, and the phase-matching of the two conditions gives the pole-zero offset $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$ exactly — hence $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the gate threshold. (The turning-point atom feeding this analysis, the rational envelope $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2}) \in [1, \frac{3}{2}]$ of Lemma 25, is formally verified in Lean 4.) The special-function obstacle that previously separated U from V is gone: the two series stand together, on the same self-contained analysis.

Section 7 records an independent arithmetic route: by Christol’s theorem, if $(u_n \bmod p)$ is not p -automatic for one prime p then U is transcendental over $\mathbb{Q}(x)$; the p -kernel is maximally non-automatic at $p = 3, 5$.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q -difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q ,

$$\Sigma_1(q) = \sum_{i \geq 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!} \rho_{i-1}, \quad S_e(q) = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q; q)_{2j}}, \quad (1)$$

and their companions Σ_0, S_0 , where $\rho_j = q^j \eta_j$ with $\eta_j = e^{-B_j}$ a Gaussian-decaying form factor. Both V and U are rational functions of these blocks over $\mathbb{Q}(x)$; concretely the relaxed travel resolvent is $G_0^T = \Sigma_0/(1 - \Sigma_1)$, and U inherits its denominator $1 - \Sigma_1$ (Proposition 1). The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \quad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$.

Proposition 1 (Invariance of the travel block). *The denominator $1 - \Sigma_1$ is the identical, cycle-marker-free object in the generating functions of U and of V ; hence U and V share the travel poles q_m .*

The proof is a strand-walk argument (every pure-travel element has cycle count 0, so the two length gradings agree on it); see the supplement.

3 Unconditional transcendence of the blocks

Theorem 2. *Each of $\Sigma_0, \Sigma_1, S_0, S_1$ is transcendental over $\mathbb{Q}(q)$.*

Proof. Each block is an integer power series with radius of convergence exactly 1. Along the squares its coefficients grow super-polynomially: for Σ_1 one has $|[q^{(j+1)^2}] \Sigma_1| \geq 2^{j+1}$, from the j^2 -gapped catalytic telescoping (the block’s coefficients $2, -4, 14, -52, 178, -856, 4626, \dots$ at $q^{(j+1)^2}$, and the bound, are reproduced from the block’s A/C recursion and machine-checked for $j \leq 9$ in Lean 4, `PolyaCarlson.lean`, theorem `coeff_bound`). A power series with integer coefficients and radius 1 is, by the Pólya–Carlson theorem [4], either a rational function whose poles are roots of unity or has the unit circle as a natural boundary. Super-polynomial coefficient growth rules out the rational alternative, so $|q| = 1$ is a natural boundary; a function with a natural boundary is transcendental over $\mathbb{Q}(q)$. $\square$

This is the unconditional core: it uses no analytic gate. It does not, however, transfer directly to V or U , which are *ratios* of blocks in which the boundary singularities can cancel; those require the pole-accumulation argument below.

4 The steepest-descent estimate

The one analytic input for both V and U is the smallness of an auxiliary integral. Let $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n \geq 1} \varphi_n y^{2n}$ with $\varphi_n = \frac{(-1)^{n+1} \zeta(2n)}{n(2\pi)^{2n}}$, and set $B_s = \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$, where $P_n(M) = \sum_{m=1}^M m^{2n}$ is the Faulhaber sum. The block S_e decomposes as $S_e = \cos W - T_2$ with $W = w e^{-\tau/2}$ and

$$T_2 = \sum_{n \geq 1} (-1)^n g_n \frac{W^{2n}}{\Gamma(2n+1)}, \quad g_s = 1 - e^{-B_s}. \quad (2)$$

Everything hinges on $T_2 \rightarrow 0$, and on its sharp size $|T_2| = O(\sqrt{\tau})$.

Lemma 3 (amplitude regularity). *On the strip $\mathcal{S} = \{0 \leq \Im s \leq W/2, \operatorname{Re} s \geq 0\}$ one has, for $\tau < \pi^2/8$,*

$$\operatorname{Re} B_s \geq -\frac{\sqrt{2}}{18} \sqrt{\tau} - 0.8 \tau^{3/2},$$

so $A := 1 - e^{-B_s}$ is bounded ($|A| \leq 2.05$) and of bounded variation on $\mathcal{S}$.

Proof sketch. Since $W/2 < \pi/\tau$ there is no resonance: writing B_s through the Γ -function form, the divergent $-\pi^2 k/\tau$ Stirling terms cancel exactly between the s -shifted and boundary pairs, so B_s is finite on $\mathcal{S}$. The lower bound is set by the $n = 1$ term $\frac{1}{36} \tau^2 \operatorname{Re}(4s^3 + 3s^2 - s)$, which minimises on $\mathcal{S}$ at the diagonal $s = \frac{W}{2}(1+i)$ to $-\frac{\sqrt{2}}{18} \sqrt{\tau} + \frac{5\sqrt{2}}{72} \tau^{3/2} \geq -\frac{\sqrt{2}}{18} \sqrt{\tau}$ (the factor $e^{-3\tau/2}$ in W makes the subleading correction *positive*). The tail is controlled by an explicit geometric majorant: for $n \geq 2$, $|\varphi_n| \leq \zeta(4)/(n(2\pi)^{2n})$ and $|P_n(2s)| \leq 2|2s|^{2n+1}/(2n+1)$ (the Bernoulli-polynomial bound $|B_{2n+1}(z)| \leq 2|z|^{2n+1}$, $|z| \geq 2n+1$), so on $|s| \leq 2W$

$$\left| \sum_{n \geq 2} \varphi_n \tau^{2n} (P_n(2s) - s) \right| \leq \sum_{n \geq 2} \frac{2\zeta(4)}{n(2n+1)} |2s| \left(\frac{|s|\tau}{\pi} \right)^{2n} \leq 0.8 \tau^{3/2},$$

a convergent geometric series in $(|s|\tau/\pi)^2 \leq 8\tau/\pi^2 < 1$; for $|s| > 2W$ the real-axis growth of $\log \Gamma(2s+1)$ forces $\operatorname{Re} B_s > 0$. Bounded variation follows because differentiating the majorant term-by-term multiplies the n -th term by $O(n)$ and preserves convergence. Full constants are verified in the supplement (`1emBbounded_tail.py`). $\square$

Rather than cite the general steepest-descent theorem, we prove the estimate directly; the argument exhibits the saddle explicitly and uses only Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the bounded-variation bound of Lemma 3 (it reproves, for this integrand, the specialization of [13, Ch. 4, Thm. 7.1]).

Lemma 4 (decay of T_2). *As $q \rightarrow 1^-$, $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin w + O(\tau)$; in particular $|T_2| = O(\sqrt{\tau}) \rightarrow 0$.*

Self-contained proof by steepest descent. Since $\cos W = \sum_{n \geq 0} (-1)^n W^{2n}/(2n)!$, (2) gives the exact alternating sum $S_e = \cos W - T_2 = \sum_{n \geq 0} (-1)^n e^{-B_n} W^{2n}/(2n)!$ (with $B_0 = 0$). Let $b(z) = e^{-B_z} W^{2z}/\Gamma(2z+1)$, analytic for $\operatorname{Re} z > -\frac{1}{2}$ (each term of B_z is a polynomial in z). As $\pi \csc \pi z$ has residue $(-1)^n$ at each integer, Cauchy's theorem gives $S_e = \frac{1}{2\pi i} \oint b(z) \pi \csc(\pi z) dz$ over a contour enclosing $\{0, 1, 2, \dots\}$. Write the integrand as $e^{h(z)} \pi \csc \pi z$ with $h(z) = 2z \log W - \log \Gamma(2z+1)$; including the $\csc$ phase, the exponent is stationary where $2 \log W - 2\psi(2z+1) \pm i\pi = 0$, so by $\psi(w) = \log w + O(1/w)$ there are two simple, non-degenerate saddles

$$z_{\pm}^* = \pm \frac{i}{2} W(1 + O(1/W)), \quad h''(z_{\pm}^*) = \pm \frac{4i}{W} + O(W^{-2}) \neq 0.$$

Deform the contour to the steepest-descent paths through $z_{\pm}^*$; this is legitimate because b decays super-exponentially as $\operatorname{Re} z \rightarrow +\infty$ and $\csc$ suppresses $|\Im z| \rightarrow \infty$. A direct Stirling evaluation shows that at the modulation-free level $B \equiv 0$ the two saddles sum to $\cos W$ *exactly* — the divergent factors $e^{\pm\pi W/2}$ from $\csc(\pi z_{\pm}^*)$ and from $1/\Gamma(\pm iW + 1)$ cancel identically, each saddle contributing $\frac{1}{2}e^{\pm iW}$. Restoring the modulation scales each saddle by $e^{-B_{z_{\pm}^*}}$; since $P_1(2z) = \frac{2z(2z+1)(4z+1)}{6}$ and $\varphi_1 = \frac{1}{24}$,

$$B_{z_+^*} = \varphi_1 \tau^2 (P_1(iW) - \frac{iW}{2}) + O(\tau^{3/2}) = \frac{\tau^2}{24} \left(-\frac{i}{3} W^3 \right) + O(\tau^{3/2}) = -i \frac{\sqrt{2}}{36} \sqrt{\tau} + O(\tau)$$

(using $\tau^2 W^3 = 2\sqrt{2}\sqrt{\tau}$, and $B_{z_-^*} = \overline{B_{z_+^*}}$. Hence

$$S_e = \frac{1}{2} e^{-B_{z_+^*}} e^{iW} + \frac{1}{2} e^{-B_{z_-^*}} e^{-iW} + E = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + E.$$

The remainder E is $O(\tau)$: parametrising the descent path by arclength s , the integrand equals $e^{h(z^*)} e^{-s^2} (1 + O(sW^{-1/2}) + O(\tau))$, and term-by-term Gaussian integration (the odd powers vanishing) leaves an error controlled by the next saddle coefficient $\propto h'''(z^*) = O(W^{-2}) = O(\tau)$, the total variation of e^{-B_z} along the path being bounded by Lemma 3. The saddle prediction matches S_e to $O(\tau)$ over $\tau = 0.02, 0.01, 0.005$ numerically (`saddle_verify.py`). Thus $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau)$, with $\sin W = \sin w + O(\sqrt{\tau})$. $\square$

Remark 5 (the Hubbard–Stratonovich integral representation). Writing $q^{j(j+1)} = q^j e^{-j^2 \tau}$ and linearising the Gaussian through $e^{-j^2 \tau} = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du$, the denominator block admits the *exact* real integral

$$S_e = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} e^{-u^2/(4\tau)} \Psi(u, q) du, \quad \Psi(u, q) = \sum_{j \geq 0} \frac{(-2(1-q)q e^{iu})^j}{(q; q)_{2j}}, \quad (3)$$

the interchange of sum and integral being justified by absolute convergence ($\sum_j |Z|^j / (q; q)_{2j} < \infty$, $Z = -2(1-q)q e^{iu}$, against the integrable Gaussian). Because the coefficients of Ψ in $\zeta = e^{iu}$ are real, $\Psi(-u, q) = \overline{\Psi(u, q)}$, so (3) is real and may be written over a single variable, $S_e = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} \operatorname{Re} \Psi du$; the exact Gaussian moment underlying it, $\int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du = \sqrt{4\pi\tau} e^{-\tau j^2}$, is machine-checked in Lean 4 (`GaussHS.lean`, theorem `gaussian_moment`; axioms `propext`, `Classical.choice`, `Quot.sound`, no sorry).

We caution that this representation, though exact and convenient, is *not* an analytic shortcut — a point where an earlier draft overreached. The amplitude $\Psi(\cdot, q)$ is entire in ζ , but it is *not* uniformly bounded as $q \rightarrow 1^-$: on any fixed window it grows like $\cosh(w \sin(u/2))$ (numerically $|\Psi|$ already exceeds 300 at $u = 1$, $\tau = 10^{-2}$), and its termwise limit $\cos(w e^{iu/2})$ differs from Ψ by an $O(1)$ factor at the dominant scale $j \sim \tau^{-1/2}$ (where the subleading part of $(q; q)_{2j}$ is itself $O(1)$; numerically $\Psi / \cos(w e^{iu/2}) \approx 0.62$). The size of S_e is therefore governed by the balance between this exponential growth and the Gaussian — a simple *complex* saddle at $u^* \approx -\sqrt{2\tau}$, of the same order of difficulty as the direct sum, *not* a bounded-amplitude stationary-phase integral with trivial hypotheses. In particular the bounded-variation input of Lemma 3 is *not* eliminated by passing to (3); it, or the equivalent simple-saddle estimate of Lemma 4, remains the analytic core. The value of (3) is representational (one real variable, an entire integrand, an exact Lean-checked moment), not a reduction of hypotheses; the numerical agreement of its saddle value with S_e is recorded in the supplement (`gaussint_verify.py`).

Remark 6 (an elementary derivation of the coefficients of Lemma 4). The leading term and the constant $\sqrt{2}/36$ of Lemma 4 can be obtained without the steepest-descent theorem, by an exact algebraic reduction. Writing $1 - e^{-k\tau} = 2e^{-k\tau/2} \sinh(k\tau/2)$ throughout, the j -th term of $S_e = \sum_{j \geq 0} (-2(1-q))^j q^{j(j+1)} / (q; q)_{2j}$ collapses to

$$S_e = \sum_{j \geq 0} (-1)^j e^{-j\tau} \frac{\sinh^j(\tau/2)}{\prod_{k=1}^{2j} \sinh(k\tau/2)} = \sum_{j \geq 0} (-1)^j \frac{(2/\tau)^j}{(2j)!} e^{-j\tau + E_j}, \quad E_j = \frac{j\tau^2}{24} - \frac{\tau^2}{144} (2j)(2j+1)(4j+1) + O(\tau^4 j^5),$$

E_j being the exact $\sinh$ -curvature defect. Two elementary facts finish the extraction. First, the factor $e^{-j\tau}$ resums *in closed form*: $\sum_j (-1)^j (2e^{-\tau}/\tau)^j / (2j)! = \cos W$ with $W = \sqrt{2/\tau} e^{-\tau/2}$, which is exactly the frequency of Lemma 4 — so the phase correction needs no estimate at all. Second, the entire-function moment identity $\sum_j (-1)^j y^j j^m / (2j)! = (y d/dy)^m \cos \sqrt{y}$ evaluates the curvature term through its leading part $-\frac{\tau^2}{9} (y d/dy)^3 \cos \sqrt{y} = -\frac{\tau^2}{9} \cdot \frac{W^3}{8} \sin W = -\frac{\sqrt{2}}{36} \sqrt{\tau} \sin W$ (using $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau}$), giving

$$S_e = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau), \quad W = \sqrt{2/\tau} e^{-\tau/2},$$

all constants explicit and verified to ten digits (`elemcos_verify.py`). Carried to higher moments the same reduction produces every subsequent coefficient as a finite combination of $W^k \sin W$, $W^k \cos W$, so it serves the $O(\tau^2)$ -relative demand of the U -numerator in the same elementary terms. The one input it does *not* render elementary is the remainder estimate — that the alternating expansion is asymptotic with $O(\tau)$ error. That step is a *cancellation* statement: not only does the triangle inequality on $\sum_j (2/\tau)^j / (2j)! |\cdot|$ return $\cosh W \asymp e^{\sqrt{2/\tau}}$, but the partial sums of the $\cos W$ series themselves swing up to $\asymp e^{c\sqrt{2/\tau}}$ before cancelling to $O(1)$ (numerically $\max_J |S_J| = 9, 48, 1459, 7.6 \times 10^4$ at $\tau = 0.1, 0.05, 0.02, 0.01$). Hence *every* partial-summation method — Abel, summation-by-parts, Euler transformation — fails, since each is controlled by these exponentially large partial sums; only a global resummation captures the cancellation. This global resummation is exactly the two-saddle steepest-descent computation carried out self-contained in the proof of Lemma 4: the saddles $z_{\pm}^* = \pm iW/2$ supply the cancellation the partial sums cannot. Together, the elementary extraction here and that saddle argument render Lemma 4 independent of any cited asymptotic theorem, resting only on Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the bounded-variation bound of Lemma 3.

Remark 7 (derivation of the U -numerator amplitude). The single U -side input — the bound $|\sum_k d_k| \leq \frac{3}{8\sqrt{2}} \tau^{5/2}$ at the travel poles on the one explicit series $P_{12} = \frac{2q^3}{1-q^3} \sum_k d_k$, $d_k = (-2)^k (1-q)^k q^{k^2+3k} / [(q^2; q^2)_k (q^5; q^2)_k]$ — submits to the identical reduction. Writing $1-q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$ throughout collapses the terms *exactly* to

$$d_k = (-1)^k e^{-k\tau} \frac{\sinh^{k+1}(\frac{\tau}{2}) \sinh(\frac{3\tau}{2}) \sinh((k+1)\tau)}{\prod_{m=1}^{2k+3} \sinh(\frac{m\tau}{2})},$$

and the elementary expansion (exact linear absorption into the argument, Faulhaber sums for the $\sinh$ -curvature, the moment identities $k^m \mapsto (y d/dy)^m$) yields, with $\Phi(y) = (\sin W - W \cos W) / (2W^3)$, $W = \sqrt{y}$, $D = y d/dy$,

$$\sum_k d_k = 6 \left[\Phi + c_3 D^3 \Phi + c_2 D^2 \Phi + \frac{c_2^2}{2} D^6 \Phi + (c_5 + c_2 c_3) D^5 \Phi + \frac{c_3^3}{6} D^9 \Phi \right] (y_*) + O(\tau^3),$$

$y_* = \frac{2}{\tau} e^{-\tau-23\tau^2/36}$, $c_2 = -\frac{5}{12} \tau^2$, $c_3 = -\frac{1}{9} \tau^2$, $c_5 = \frac{1}{450} \tau^4$ — *every constant an exact rational, derived, not fitted* (`elemY3_verify.py`; the error is verified $O(\tau^3)$ over $\tau \in [0.005, 0.08]$ and the formula

reproduces the exact series at the poles $m = 6, \dots, 32$ to the displayed accuracy). At the travel poles the expansion gives $|\sum_k d_k|/\tau^{5/2} \rightarrow 3/(8\sqrt{2}) = 0.2652$, i.e. exactly the observed gate value $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ with its fourfold margin: structurally, the pole sits at offset δw from the zero of the displayed expansion, and the gate constant is $(3\tau/2) \delta w/\tau^{5/2}$.

The *phase-matching step* that fixes δw is carried out by the same machinery. The pole equation $\Sigma_1(q_m) = 1$ collapses exactly to $1 - \Sigma_1 = \sum_{j \geq 0} (-1)^j \sinh^j(\frac{\tau}{2}) / \prod_{m=1}^{2j} \sinh(\frac{m\tau}{2})$ — the S_e -series stripped of its $e^{-j\tau}$ phase — so the identical expansion applies with $y_* = \frac{2}{\tau} e^{\tau^2/36}$, $c_2 = -\frac{1}{12}\tau^2$ and the *universal* $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$. Solving the two phase conditions perturbatively in the common variable $w = \sqrt{2/\tau}$ near each grid point $(M + \frac{1}{2})\pi$ gives

$$\chi_{\text{pole}} = -\frac{\sqrt{2}}{36}\sqrt{\tau} + \frac{41\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2), \quad \chi_{\text{zero}} = -\frac{\sqrt{2}}{36}\sqrt{\tau} - \frac{259\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2).$$

The $\sqrt{\tau}$ phases agree *identically* — both are produced by the universal cubic curvature c_3 , and the constant $\sqrt{2}/36$ is that of Lemma 4; this is precisely why the travel poles track the zeros of $Y_3(1)$. The $\tau^{3/2}$ terms differ by exactly

$$\delta w = \chi_{\text{pole}} - \chi_{\text{zero}} = \frac{\sqrt{2}}{8}\tau^{3/2} + O(\tau^2), \quad \text{whence} \quad \left| \sum_k d_k \right| = \frac{3\tau}{2} \delta w (1 + O(\sqrt{\tau})) = \frac{3}{8\sqrt{2}}\tau^{5/2}(1 + O(\sqrt{\tau})),$$

and therefore $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.17678 < 1/\sqrt{2}$: *the numerator gate, derived, with its fourfold margin*. Verification: the two phase series match the exact pole and zero locations to seven digits at $m = 10, 16, 22, 28$, and $\delta w/\tau^{3/2} = 0.17699, 0.17687, 0.17683, 0.17681 \rightarrow \sqrt{2}/8 = 0.176777$ (`phase_match_verify.py`); the pole-side series independently reproduces the McMahon-type coefficients of the travel-pole expansion. The remainder discipline throughout is that of Lemma 4: every coefficient extracted elementarily, the $O(\cdot)$ terms controlled by the same two-saddle argument.

5 The relaxed series V

Theorem 8. *The relaxed growth series V is transcendental over $\mathbb{Q}(x)$.*

Proof. V contains the travel resolvent $(1 - \Sigma_1)^{-1}$ times a bulk-dressing factor holomorphic and non-vanishing at the q_m ; thus V has a genuine pole at each travel pole q_m , and the $q_m \uparrow 1$ accumulate at the boundary of the disc of convergence. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$; an infinite pole set accumulating at the boundary is impossible for an algebraic function, so V is transcendental — *provided* the poles are not cancelled by the numerator. Non-cancellation is exactly the statement that $S_e(q_m) = \cos W - T_2 \neq 0$ for all large m , i.e. that $|T_2| < |\cos W|$ there; since $|\cos W| \asymp \sqrt{\tau/2}$ at the poles while $|T_2| = O(\sqrt{\tau})$ with the small constant $\sqrt{2}/36$ (Lemma 4), the inequality holds with a fixed margin. Hence V is transcendental; its sole analytic input is the simple-saddle estimate of Lemma 4, which we prove here from first principles (Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the explicit bounded-variation bound of Lemma 3), rather than by appeal to a general theorem. $\square$

Corollary 9. *The exponential growth rate of both u_n and v_n is $\beta_2 = 1.4916177871\dots$, the reciprocal of the smallest travel pole; the value $\frac{3}{2}$ is excluded. This is unconditional (it needs only the location of the dominant travel pole, not Lemma 4).*

Remark 10 (the number β_2 : a value-transcendence question). Whether the number β_2 is transcendental is open, but the question has a clean reduction. Since $\beta_2 = 1/\sqrt{q^*}$ with $\Sigma_1^T(q^*) = 1$, the

number β_2 is algebraic iff q^* is; if q^* were algebraic, the series Σ_1^T would take the *rational* value 1 at an algebraic interior point of its disc. For the Jacobi-theta class this is impossible: by Nesterenko's theorem on Eisenstein values [11], $\sum_n \alpha^{n^2}$ is transcendental at *every* algebraic α with $0 < |\alpha| < 1$ [8]. The blocks here are theta-*like* — their coefficients ride the squares $q^{(j+1)^2}$, which is the source of the natural boundary of Theorem 2 — but their Euler-product denominators $(q; q)_{2k+1}$ spread the support densely, and no analogue of [11] is known off the modular class.

Concretely the reduction localizes on a single confluent basic hypergeometric function. Writing $\Sigma_1^T(q) = 1 - S(q)$,

$$S(q) = \sum_{k \geq 0} \frac{(-2(1-q))^k q^{k^2}}{(q; q)_{2k}} = {}_1\phi_1(0; q; q^2, 2q(1-q)),$$

a q -Bessel-class function (using $(q; q)_{2k} = (q; q^2)_k (q^2; q^2)_k$); β_2 is algebraic iff the least positive zero $q^* = 0.449453630\dots$ of S is. We record what the natural tools give here, since it locates the difficulty precisely. First, S is *neither* q -holonomic *nor* a Mahler function in q (verified to $O(q^{48})$): the diagonal specialization $z = 2q(1-q)$ ties the argument to the base and destroys the single-variable q -difference structure, so the methods of Mahler–Nishioka and of holonomy do not apply. Second, two elementary refinements fail *decisively*, not for want of effort: an ultrametric argument is void — the v -adic and archimedean sums of $S(\alpha)$ are unrelated numbers, so $|S(\alpha)|_v = 1$ does not bound the real value — and a Liouville/height bound is beaten by the q -Pochhammer denominators, whose growth over the conjugates of modulus > 1 is cubic ($\sim K^3$) against the merely quadratic ($\sim K^2$) tail, Kronecker-sharply (equality would need Mahler measure 1, i.e. a root of unity). A more promising reformulation uses two variables. Set $G(q, z) = {}_1\phi_1(0; q; q^2, z) = \sum_k (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$, so $S(q) = G(q, 2q(1-q))$; unlike S , the function G is q -holonomic in z , satisfying the second-order q -difference equation

$$q G(q, z) - (q + 1 - qz) G(q, q^2 z) + G(q, q^4 z) = 0.$$

One checks that $z^* := 2q^*(1 - q^*) = 0.4948901\dots$ is the *smallest positive zero* of $G(q^*, \cdot)$. Hence β_2 algebraic would force an *algebraic zero of a q -Bessel-class function at an algebraic base* — and the classical analogue is false: the zeros of the Bessel function J_ν are transcendental (Siegel). A q -analogue of Siegel's theorem for the zeros of ${}_1\phi_1$ would therefore settle β_2 . The right framework is arithmetic. Writing the equation as a system $Y(q^2 z) = A(z)Y(z)$ with $Y = (G(z), G(q^2 z))^T$, the companion matrix

$$A(z) = \begin{pmatrix} 0 & 1 \\ -q & q + 1 - qz \end{pmatrix}$$

is *polynomial* in z with $\mathbb{Z}[q]$ entries and $\det A = q$, so the iterated matrices carry no denominators and the Casoratian obeys $W(q^2 z) = qW(z)$, i.e. $W(z) = cz^{1/2}$: the two solutions form a q -Bessel pair with Frobenius exponents $0, \frac{1}{2}$. The second solution is explicit, $\tilde{G}(z) = z^{1/2}H(z)$ with $H(z) = \sum_k (-1)^k q^{k^2} (1-q)z^k / (q; q)_{2k+1}$, and the pair satisfies the exact q -Wronskian identity

$$q G(z) H(q^2 z) - G(q^2 z) H(z) = q - 1,$$

so at the zero z^* one has the clean relation $G(q^2 z^*)H(z^*) = 1 - q$. Moreover $G(q, \cdot)$ is *entire of order zero* — a q - E -function, not a q - G -function. Hence β_2 transcendental would follow from a q -analogue of the Siegel–Shidlovskii theorem for this q - E -function: such a theorem gives $G(q, \alpha)$ transcendental at every algebraic $\alpha \neq 0$ off the singular locus, whereas $G(q^*, z^*) = 0$ is algebraic — forcing q^* , and with it β_2 , transcendental (and by the shared q -connection core, likewise U). All

the structural hypotheses hold here (polynomial companion matrix, constant determinant, entire order-zero solution); the sole open point is whether the available q -Siegel–Shidlovskii results (André; Di Vizio–Hardouin) extend to a ${}_1\phi_1$ that is *irregular* at $z = \infty$ — a precise, expert-checkable question, not an absence of theory. (Note the contrast with the *function-level* results of this paper, which say nothing about individual interior values.)

For *irrationality* (weaker than transcendence) the classical architecture transports verbatim. G and H are Hahn–Exton (third Jackson) q -Bessel functions: $G(q, z) \propto J_{-1/2}(\sqrt{z}/q; q^2)$ and $H(q, z) \propto J_{+1/2}(\sqrt{z}/q; q^2)$ (verified; the zeros correspond exactly), i.e. the q -analogues of $\cos$ and $\sin$ — and z^* is the q -analogue of $(\pi/2)^2$. Niven’s irrationality proof for π runs on the Lommel-polynomial ladder $J_{m+1/2}(r) = R_{m,1/2}(r)J_{1/2}(r) - R_{m-1,3/2}(r)J_{-1/2}(r)$ evaluated at the zero of $J_{-1/2} = \sqrt{2/\pi r} \cos$; the exact q -analogue of every ingredient exists (Koelink–Swarttouw): the ladder $J_{\nu+1}(x; q^2) = ((1 - q^{2\nu})/x + x)J_{\nu} - J_{\nu-1}$, the q -Lommel connection $J_{\nu+m} = R_{m,\nu}J_{\nu} - R_{m-1,\nu+1}J_{\nu-1}$ with $x^m R_{m,\nu} \in \mathbb{Z}[q, x^2]$, and the second-kind functions carrying $q^{m^2/2}$ prefactors — all verified numerically for our pair. Moreover, since a rational zero would make $G(q^*, z^*) = 0$ a *rational value* at rational base and argument, β_2 is irrational as soon as this Hahn–Exton function omits the value 0 at rational points — a plain *value-irrationality* statement, strictly weaker than every transcendence target above. What blocks even this is now a single quantified invariant: the *arithmetic ratio* ρ of the series (b -adic decay exponent of the terms over b -adic growth exponent of the cleared denominators, $q = s/t$ in lowest terms). Every proven q -irrationality theorem (Tschakaloff, q -exponential, the Bézivin–Bundschuh–Väänänen classes) lives at $\rho \geq 1$, where the Pochhammer denominator itself supplies the decay; our G has an independent q^{k^2} numerator against the double Pochhammer $(q; q)_{2k}$, giving $\rho = \frac{1}{2}$. All three classical constructions fail by exactly this margin, measured: the z -Padé remainder $q^{3.5n^2}$ against height q -degree $\approx 9n^2$; the z -direction continued fraction (convergents in $\mathbb{Z}[q, z]$, Pincherle limit the H -ratio) converges only geometrically against t^{n^2} heights; and the ν -ladder at the zero has $J_{m+1/2}(x_0) = R_{m,1/2}(x_0)J_{1/2}(x_0) \sim Cx_0^m$ — geometric, because the q -factorial is geometric where Niven’s $m!$ is super-exponential — against t^{m^2} . The ν -ladder is in fact fully costed: the q -Lommel coefficients satisfy $\deg_q p_m = m^2$ *exactly* (immediate from the recurrence, verified symbolically), so their denominators at $q = s/t$ are the pure power t^{m^2} — carrying *no* cyclotomic structure, so the lcm-compression that powers the little- q -Legendre proofs has nothing to act on — while by Poincaré–Perron the m -recurrence, whose coefficient tends to $x_0 + 1/x_0$ with *distinct* characteristic roots $x_0, 1/x_0$, admits *only geometric* solutions: no theta-type q^{m^2} -decaying solution exists in the ν -direction for either kind (the second kind’s $q^{\nu^2/2}$ prefactor is exactly cancelled by the $(q^{-\nu+1}; q)_{\infty}$ blow-up of its shifted argument). Hence the cleared ν -ladder forms grow like $t^{m^2}x_0^{-m} \rightarrow \infty$ for *every* rational $q = s/t$, including $s = 1$. The $b \leftrightarrow z$ transformation ${}_1\phi_1(0; q; q^2, z) = \frac{(z; q^2)_{\infty}}{(q; q^2)_{\infty}} {}_1\phi_1(0; z; q^2, q)$ redistributes but conserves ρ . The structural summary: the theta-type decay lives only in the z -direction, where the equation is *irregular* at $z = \infty$ (Newton polygon slopes $\{0, 1\}$) and every integral structure carries the double-Pochhammer weight $\rho = \frac{1}{2}$; the ν -direction is *regular* (Poincaré class) with clean $\mathbb{Z}[q, X]$ arithmetic but only geometric decay. Decay and arithmetic live in different directions and never meet — the same irregularity that obstructs q -Siegel–Shidlovskii creates the arithmetic deficit. The precise open door remains a q -irrationality criterion at $\rho = \frac{1}{2}$ — one notch beyond the proven $\rho \geq 1$ theory.

The failure pattern above is not an artifact of particular constructions; it is forced by exact arithmetic facts, which we record as a theorem-level ledger.

Proposition 11 (Arithmetic no-go ledger at $\rho = \frac{1}{2}$). *Let $q = s/t \in (0, 1)$ with $\gcd(s, t) = 1$, and let $g_k = (-1)^k q^{k(k-1)}/(q; q)_{2k}$ be the Taylor coefficients of $G(q, \cdot)$. Then:*

- (i) (exact denominators) *in lowest terms $g_k = (-1)^k s^{k^2-k} t^{k^2+2k} / \prod_{i=1}^{2k} (t^i - s^i)$; in particular*

$\text{den}(g_k) = \prod_{i=1}^{2k} (t^i - s^i) = t^{k(2k+1)(1+o(1))}$ exactly, while $|g_k| = (s/t)^{k^2(1+o(1))}$, so the arithmetic ratio is $\rho = \log(t/s)/(2 \log t) \leq \frac{1}{2}$, with equality iff $s = 1$.

- (ii) (ν -ladder arithmetic) the q -Lommel polynomials of Koelink–Swarttouw satisfy $\deg_q p_m = m^2$ exactly, with leading q -coefficient $(-1)^m$; hence at $q = s/t$ their denominators are the pure power t^{m^2} , carrying no cyclotomic structure.
- (iii) (ν -direction regularity) the ladder recurrence $u_{m+1} = ((1 - q^{2m+1})/x_0 + x_0)u_m - u_{m-1}$ has coefficients converging exponentially to $x_0 + 1/x_0$ with distinct characteristic roots $x_0^{\pm 1}$; by Poincaré–Perron every nonzero solution satisfies $|u_m|^{1/m} \rightarrow x_0^{\pm 1}$. No theta-type (q^{cm^2} -decaying) solution exists in the ν -direction, for either kind. Consequently the cleared ν -ladder forms are $\gg t^{m^2} x_0^{-m} \rightarrow \infty$ for every rational q , including $s = 1$.
- (iv) (cyclotomic floor) any linear form built over the coefficients $g_0, \dots, g_M$ has decay at best $q^{M^2(1+o(1))}$, while its cyclotomic support is $\{\Phi_d(t, s) : d \leq 2M\}$; even if every Φ_d appeared at most once — the idealized best case of any lcm/acceleration argument — the denominator exceeds $t^{\sum_{d \leq 2M} \varphi(d)} = t^{(12/\pi^2 + o(1))M^2}$, and $12/\pi^2 = 1.2159\dots > 1$. Hence no cyclotomic-compression (Rhin–Viola-type) refinement of the Padé, continued-fraction, or ladder families can produce integer forms tending to 0. Moreover the true solved Padé denominators contain growing non-cyclotomic irreducible factors (at $n = 3$, one of degree 26 with coefficients up to 4; no integer d has $\varphi(d) = 26$), which no lattice-of-cyclotomes argument addresses.

Proof. (i) $q^{k(k-1)} = s^{k(k-1)}/t^{k(k-1)}$ and $1/(q; q)_{2k} = t^{k(2k+1)}/\prod_{i \leq 2k} (t^i - s^i)$; the numerator $s^{k^2-k} t^{k^2+2k}$ is coprime to each factor since $\gcd(st, t^i - s^i) = 1$. (ii) Induction: $p_{m+1} = (X + 1 - q^{2m+1})p_m - Xp_{m-1}$, and the top q -degree term of p_{m+1} is $-q^{2m+1} \cdot (\text{top of } p_m)$, of degree $m^2 + 2m + 1 = (m+1)^2$, which cannot cancel against Xp_{m-1} (degree $(m-1)^2$). (iii) The perturbation $(1 - q^{2m+1})/x_0 - 1/x_0$ decays geometrically, so the Poincaré–Perron theorem applies with the limit equation $u_{m+1} = (x_0 + 1/x_0)u_m - u_{m-1}$, whose roots $x_0 \neq 1/x_0$ are simple. (iv) A form using $g_0, \dots, g_M$ vanishes to order at most M at $z = 0$, so its value at a fixed z_0 inside the disc is $\gg |g_{M+1} z_0^{M+1}| = q^{M^2(1+o(1))}$ in the truncation-based families; its denominators involve $(t^i - s^i)$ for $i \leq 2M$, i.e. $\Phi_d(t, s)$ for $d \leq 2M$, and $\sum_{d \leq N} \varphi(d) = \frac{3}{\pi^2} N^2 + O(N \log N)$ gives the floor with $N = 2M$. The degree-26 factor is exhibited by the exact $n = 3$ computation (`code/zeta_probe/route_b/beta2_qniven_ladder.py`). $\square$

Remark 12 (scope). Proposition 11 closes every construction whose coefficients live in the natural $\mathbb{Z}[q^{\pm 1}, z]$ -lattices of the q -difference module — Taylor Padé and Hermite–Padé, the argument-direction continued fraction, the ν -ladders of both kinds — together with all cyclotomic-compression refinements of them. It does *not* exclude a fundamentally different family of forms; producing one is exactly what a $\rho = \frac{1}{2}$ criterion would mean. The ledger thereby locates β_2 ’s arithmetic at a precise frontier of q -Diophantine approximation.

The remaining freedom can be narrowed further: the finite places of $\mathbb{Q}$ are provably useless here. The series $S(q) = \sum_k g_k z^{*k}$, $z^* = 2q(1 - q)$, has rational terms, so besides its real sum it possesses s -adic and t -adic *avatars* — the same series summed in $\mathbb{Q}_s$ and $\mathbb{Q}_t$, where it also converges (the valuations of the terms tend to $+\infty$).

Lemma 13 (adelic units). *Let $q = s/t \in (0, 1)$, $\gcd(s, t) = 1$, and $z^* = 2q(1 - q)$. For every $k \geq 1$ and every prime p : if $p \mid t$ with $e = v_p(t)$, then*

$$v_p(g_k z^{*k}) = k^2 e + k[p = 2] \geq k^2,$$

(exact, verified symbolically), and if $p \mid s$ then $v_p(g_k z^{*k}) \geq k^2$. Consequently the series converges in $\mathbb{Q}_p$ for every $p \mid st$, and the finite-place avatars satisfy $S_p \equiv 1 \pmod{p^{v_p(st)}}$; in particular $S_t \equiv 1$

(mod t) and $S_s \equiv 1 \pmod{s}$: they are units, hence nonzero — regardless of whether the real avatar vanishes.

Proof. Each factor $t^i - s^i$ of $\text{den}(g_k)$ is a p -unit for every $p \mid st$ (it is $\equiv -s^i$ or $\equiv t^i \pmod{p}$), so by Proposition 11(i), $v_p(g_k) = (k^2 + 2k)e$ for $p \mid t$ and $v_p(g_k) = (k^2 - k)v_p(s)$ for $p \mid s$. From $z^* = 2s(t - s)/t^2$: for $p \mid t$, $v_p(t - s) = 0$ and $v_p(z^*) = [p = 2] - 2e$, giving $v_p(g_k z^{*k}) = (k^2 + 2k)e + k([p = 2] - 2e) = k^2e + k[p = 2]$; for $p \mid s$, $v_p(z^*) = v_p(2s) \geq v_p(s) \geq 1$, giving $v_p(g_k z^{*k}) \geq (k^2 - k) + k = k^2$. All are ≥ 1 for $k \geq 1$, so each avatar is 1 plus terms of positive valuation. $\square$

Remark 14 (exact adelic balance, and the shape of any future criterion). Lemma 13 has two consequences. First, it extends the no-go ledger from the module's coefficient lattices to the *values*: no divisibility or smallness can be sourced from the finite places, since the finite-place avatars are units — a hypothetical rationality of β_2 would be a maximally place-asymmetric event, the real avatar vanishing while every finite avatar equals a unit $\equiv 1$. Second, it makes the $\rho = \frac{1}{2}$ balance exact in the sense of the product formula: the term $g_k z^{*k}$ carries archimedean decay $(s/t)^{k^2(1+o(1))}$, s -adic and t -adic numerator content $s^{k^2}t^{k^2}$, and denominator $t^{2k^2(1+o(1))}$ — the books balance with *zero margin* at the k^2 scale, so any successful construction must be lossless at all three places simultaneously and win at the geometric scale. Every mechanism surveyed above loses a factor t^{ck^2} somewhere; a $\rho = \frac{1}{2}$ criterion is therefore forced to be a purely *archimedean cancellation* phenomenon. This suggests the precise principle that would settle β_2 , which we record as a conjecture: for the irregular rank-two q -difference module with polynomial companion matrix and $\det A = q$, the vanishing locus is *adelically rigid* — the entire function $G_v(q, \cdot)$ cannot vanish at a rational point z_0 at which all finite-place avatars are units (as at $z_0 = z^*$, by Lemma 13) at one place v of $\mathbb{Q}$ while the avatars at all other convergent places are units. The finite-place half of this rigidity is exactly Lemma 13; its archimedean half implies $\beta_2 \notin \mathbb{Q}$. This is a q -analogue, for irregular q -difference modules, of the place-uniformity enjoyed by E - and G -function values, where the exceptional sets of Siegel–Shidlovskii–André theory are independent of the place.

Further analysis of the archimedean obstruction yields several exact structural facts and a faithful reduction. All are verified numerically to 40–60 digits (`code/zeta_probe/route_b/beta2_*.py`).

Proposition 15 (structure of the zero curve). *Write $Q = q^2$ and let $0 < z_1(q) < z_2(q) < \dots$ be the positive zeros of $G(q, \cdot)$.*

- (i) (base change) $G(q, z) = \frac{(q^2; q^2)_\infty}{\sqrt{q}(q; q^2)_\infty} z^{1/4} J_{-1/2}^{(3)}(\sqrt{z}/q; q^2)$, where $J_\nu^{(3)}$ is the Hahn–Exton (third Jackson) q -Bessel function; thus G is, up to an explicit prefactor, the Hahn–Exton q -cosine, and its zeros are those of a classical, well-studied object.
- (ii) (zero lattice) $z_k(q) Q^k \rightarrow 1$ as $k \rightarrow \infty$, with theta-suppressed dressing $\delta_k := z_k Q^k - 1 = O(c^k Q^{k^2})$ (numerically $\delta_0 = -0.5051$, $\delta_1 = -1.17 \cdot 10^{-2}$, $\delta_2 = -1.03 \cdot 10^{-5}, \dots$); $z^* = z_1(q^*)$ is the maximally dressed member, displaced from the bare lattice point $Q^0 = 1$ by an $O(1)$ correction — it is not a torsion/lattice point of the connection curve.
- (iii) (power sums are rational) the symmetric functions $\sigma_p(q) := \sum_{n \geq 1} z_n(q)^{-p}$ lie in $\mathbb{Q}(q)$ for every $p \geq 1$ (Newton's identities on the $\mathbb{Q}(q)$ -coefficients g_k); e.g. $\sigma_1 = 1/((1 - q)(1 - q^2))$. Consequently $z^*(q) = \lim_{p \rightarrow \infty} \sigma_p(q)/\sigma_{p+1}(q)$ with geometric rate $(z_1/z_2)^p$, giving an all-orders $\mathbb{Q}(q)$ -approximation of the zero curve.

Two of the no-go's ingredients also sharpen. First, the *cause* of $\rho = \frac{1}{2}$ is exactly the *double* Pochhammer: $\rho = \log(t/s)/(2 \log t)$, and the factor 2 in the denominator is the $2k$ in $(q; q)_{2k}$; a single Pochhammer would give $\rho > 1$ (whence Liouville) when $t > s^2$. The intractability is intrinsic to the base- q^2 Hahn–Exton structure, not to the choice of construction. Second, the finite-place primes are *disjoint* from st : $\prod_{i \leq 2N} (t^i - s^i)$ is coprime to st (only the prime 2, via lifting-the-exponent when s, t are both odd, contributes an $O(N)$ term), so the arithmetic of every partial-sum denominator is supported off $\{s, t\}$ — orthogonal to the unit residues of Lemma 13. The Hankel/determinantal escape is likewise closed exactly: $\det(g_{i+j})_{0 \leq i, j \leq n}$ has denominator of *cubic* archimedean size ($\log |\text{den}| \sim c n^3$), one power worse than the quadratic remainder.

Finally, the Nesterenko route is blocked more precisely than by non-holonomicity alone. Nesterenko's method needs a finitely generated δ -closed ($\delta = q d/dq$) $\mathbb{C}(q)$ -algebra containing S , i.e. that S be *differentially algebraic*; non-holonomicity does not suffice (the quasimodular E_2 is non-holonomic yet δ -algebraic). One computes $\delta S = \sum_k a_k (k^2 - \frac{kq}{1-q} + L_k)$ with $L_k = \sum_{i=1}^{2k} i q^i / (1 - q^i) \rightarrow (1 - E_2)/24$; S carries *every* finite Eisenstein truncation L_k , not merely the limit, which is precisely why it escapes the modular differential ring. Whether S is differentially algebraic is itself open (no algebraic ODE of order ≤ 3 , degree ≤ 5 over 260 series terms); the Nesterenko route is blocked *modulo the differential transcendence of S* , which is itself open.

Remark 16 (reduction to a theta non-resonance). All routes converge on the same faithful reformulation: β_2 is irrational iff $G(q^*, z^*) \neq 0$ is forced by the arithmetic, and (Ramis–Sauloy–Zhang) this is a *theta non-resonance* statement — the connection ratio $R_q(z) = [P_{21} y_{\text{irr}}]/[P_{11} y_{\text{reg}}]$, an explicit θ_{q^2} -quotient, must satisfy $R_q(2q(1 - q)) \neq -1$ at every algebraic q . The evidence points toward truth: in the confluent limit $q \rightarrow 1$ this is exactly Lindemann's theorem that $\cos$ has no algebraic zero. But the reduction is not yet shown *easier*: $R_q(z) = -1$ is a functional-equation-consistent value that a nonconstant theta-quotient must attain somewhere, and on the diagonal $z = 2q(1 - q)$ the theta-arguments appear algebraically dependent, so no current independence theorem (Bertrand, Nesterenko, Schneider–Lang) forbids it. The archimedean obstruction is thus the exact mirror of the proven finite-place balance: units ($\equiv 1$) at every $p \mid st$, and a putative -1 at ∞ , with neither forbidden. The required connection-entry computation is carried out below.

Proposition 17 (the connection entries). *Write $Q = q^2$, $\theta_b(x) = \sum_{n \in \mathbb{Z}} b^{n(n-1)/2} x^n$. The equation for G has, at $z = \infty$, the formal solution basis $f_1(z) = \theta_Q(-z) \hat{g}(z)$ and $f_2(z) = h(z)/\theta_Q(-z/q)$, where $h(z) = \sum_k c_k z^{-k}$ is convergent (entire in $1/z$, $c_k = O(Q^{k^2/2})$) while $\hat{g}(z) = \sum_k b_k z^{-k}$ is q -Gevrey divergent, with the closed form*

$$b_k = \frac{q^{k-k(k-1)/2}}{(q; q)_k}$$

(equivalent to the recursion $b_k = [-(q+1)Q b_{k-1} - Q^{3-k} b_{k-2}]/(q(Q^k - 1))$; verified symbolically for $k \leq 8$). By Euler's theorem $\sum_j x^j/(q; q)_j = 1/(x; q)_\infty$, the q -Borel transform is therefore explicit:

$$\hat{B}(\xi) = \sum_{k \geq 0} b_k q^{k(k+1)/2} \xi^{-k} = \sum_{k \geq 0} \frac{(q^2/\xi)^k}{(q; q)_k} = \frac{1}{(q^2/\xi; q)_\infty},$$

meromorphic on $\mathbb{C}^*$ with poles exactly on $\xi \in q^{2+\mathbb{N}_0}$, and satisfying $\hat{B}(\xi) + \frac{q+1}{\xi} \hat{B}(q\xi) + (\xi^{-2} - 1) \hat{B}(q^2\xi) = 0$ identically (direct computation from the product form; in particular $\hat{B}(-1) = (q+1) \hat{B}(-q)$). The theta-kernel q -Laplace

$$g^{[\lambda]}(z) = \frac{\sum_{m \in \mathbb{Z}} q^{m(m-1)/2} (\lambda/z)^m \hat{B}(\lambda q^m)}{\theta_q(\lambda/z)}, \quad f_1^{[\lambda]} = \theta_Q(-z) g^{[\lambda]}(z),$$

is then singular only for $\lambda q^{\mathbb{Z}}$ meeting the pole set, i.e. for the single class $[\lambda] \in q^{\mathbb{Z}}$ of $E_q = \mathbb{C}^*/q^{\mathbb{Z}}$ (the positive-real lattice class, near which the zeros of G lie); every other class is regular. For $\lambda \notin q^{\mathbb{Z}}$ the bilateral sum converges absolutely on compacts of $\mathbb{C}^* \setminus (-\lambda q^{\mathbb{Z}})$ and $Lf_1^{[\lambda]} = 0$ exactly: the theta shifts $\theta_q(q^{-2}x) = q^{-3}x^2\theta_q(x)$ and $\theta_q(q^{-4}x) = q^{-10}x^4\theta_q(x)$ turn $f_1^{[\lambda]}(Qz)$ and $f_1^{[\lambda]}(Q^2z)$ into the same kernel sum with $\widehat{B}(\lambda q^m)$ replaced by $\widehat{B}(\lambda q^{m+2})$, $\widehat{B}(\lambda q^{m+4})$; collecting the coefficient of $q^{m(m-1)/2}(\lambda/z)^m/\theta_q(\lambda/z)$ in $Lf_1^{[\lambda]}/\theta_Q(-z)$ then yields $q[\widehat{B}(\xi) + \frac{q+1}{\xi}\widehat{B}(q\xi) + (\xi^{-2} - 1)\widehat{B}(q^2\xi)]$ at $\xi = \lambda q^m$, identically zero. (Each step was verified numerically to 40 digits; the kernel normalization was validated on convergent input to 34 digits, and $Lf_1^{[\lambda]} \approx 10^{-61}$ numerically.) Writing $G = C_1 f_1^{[\lambda]} + C_2^{[\lambda]} f_2$, numerically to 59–60 digits:

- (i) since $\det A = q$, every Casoratian $\text{Cas}(u, v)(z) := u(z)v(Qz) - u(Qz)v(z)$ of solutions satisfies $\text{Cas}(Qz) = q \text{Cas}(z)$ (immediate from $u(Q^2z) = a u(Qz) - q u(z)$), so $C_1 = \text{Cas}(G, f_2)/\text{Cas}(f_1^{[\lambda]}, f_2)$ and $C_2^{[\lambda]} = \text{Cas}(f_1^{[\lambda]}, G)/\text{Cas}(f_1^{[\lambda]}, f_2)$ are Q -elliptic — proven, and verified to 10^{-60} ;
- (ii) C_1 is a constant, independent of both z and the resummation direction λ : this is proven in Theorem 18 below (the limit argument applies along every orbit class $\neq [1]_Q$, and ellipticity extends the value); the slope-1 component of G is Stokes-invariant, and all q -Stokes ambiguity is carried by $C_2^{[\lambda]}$, nonconstant elliptic with λ^2 in its zero divisor;
- (iii) (dressing-product formula, verified to 59 digits)

$$C_1 = \frac{1}{(Q; Q)_{\infty} \prod_{k \geq 1} z_k Q^{k-1}} :$$

the Stokes-invariant connection constant is the regularized product of the zero dressings;

- (iv) at $q = q^*$ the entire apparatus evaluates: $C_1(q^*) = 2.69898382616391375\dots$, and the zero condition $G(q^*, z^*) = 0$ takes the pinned form

$$C_2^{[\lambda]}(z^*) = -C_1 \frac{f_1^{[\lambda]}(z^*)}{f_2(z^*)} = 0.011592994120624266\dots \quad (\lambda = \tfrac{3}{2}),$$

verified to 30 digits. This is the explicit instance of the theta non-resonance: every object in it is now constructed.

The constant C_1 is in fact *modular*, and this upgrades the dressing-product formula into an exact evaluation of the zero product.

Theorem 18 (the connection constant, and the zero product of the Hahn–Exton q -cosine). *For $0 < q < 1$,*

$$C_1 = \frac{1}{(q; q)_{\infty}}, \quad \text{and consequently} \quad \prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_{\infty},$$

where $z_1 < z_2 < \dots$ are the positive zeros of $G(q, \cdot)$. Likewise, for the second solution's underlying entire function H with zeros $w_1 < w_2 < \dots$, $\prod_{k \geq 1} w_k(q) q Q^{k-1} = (q^3; q^2)_{\infty}$. (Verified at three bases including q^* , to 41, 23, 26 digits for C_1 and 58 digits for the product.)

Proof. (1) *An exact identity.* From $(q; q)_{2k} = (q; q)_{\infty}/(q^{2k+1}; q)_{\infty}$ and Euler's expansion $(x; q)_{\infty} = \sum_{j \geq 0} (-1)^j q^{j(j-1)/2} x^j / (q; q)_j$ at $x = q^{2k+1}$, together with $k(k-1) + 2kj = (k+j)(k+j-1) - j(j-1)$ and the substitution $m = k+j$, absolute convergence of the double sum permits the rearrangement

$$(q; q)_{\infty} G(z) = \sum_{m \geq 0} (-1)^m q^{m(m-1)/2} z^m P_m(1/z), \quad P_m(w) := \sum_{j=0}^m c_j w^j, \quad c_j = \frac{q^{j-j(j-1)/2}}{(q; q)_j}$$

(the c_j coincide with the b_j of Proposition 17; identity checked symbolically through z^8 and numerically to 41 digits).

(2) *The constant.* Fix $u < 0$ and put $z = uQ^{-N}$. The moduli $q^{m(m-1)}|z|^m$ peak at $m_0 = N + O(1)$ and decay like $q^{(m-m_0)^2 + O(1)}$ away from it. For $1 \leq j \leq m \leq 2N$, $2Nj - j(j-1)/2 \geq Nj$, so $|c_j z^{-j}| \leq Cq^{Nj}|u|^{-j}$ and $|P_m(1/z) - 1| \leq C'q^N$ uniformly; the terms with $m > 2N$ contribute a relative $O(q^{N^2})$. The negative- m part of $\theta_Q(-z)$ is $O(|z|^{-1})$, negligible against the peak. Hence $(q; q)_\infty G(z)/\theta_Q(-z) \rightarrow 1$ along $z = uQ^{-N}$, for each fixed $u < 0$.

(3) *Geometric zero dressing.* On the circle $|z - Q^{1-k}| = \varepsilon q^k Q^{1-k}$ the identity of step (1) and the bounds of step (2) give $|(q; q)_\infty G - \theta_Q(-z)| \leq Cq^k |\theta_Q(-z)|_{\text{peak}}$, while $\theta_Q(-z)$ has a simple zero at Q^{1-k} with derivative bounded below there; Rouché gives $z_k = Q^{1-k}(1 + O(q^k))$, i.e. $\log(z_k Q^{k-1}) = O(q^k)$, absolutely summable.

(4) *The product.* $G(z) = \prod_k (1 - z/z_k)$ (order 0, genus 0, $G(0) = 1$) and $\theta_Q(-z) = (Q; Q)_\infty (z; Q)_\infty (Q/z; Q)_\infty$ (Jacobi). Along $z = -R$, $R \rightarrow \infty$,

$$\frac{G(-R)}{\theta_Q(R)} = \frac{1}{(Q; Q)_\infty (-Q/R; Q)_\infty} \prod_{k \geq 1} \frac{1 + R/z_k}{1 + RQ^{k-1}},$$

and for $x, y > 0$ the function $R \mapsto \log \frac{1+Rx}{1+Ry}$ is monotone from 0 to $\log(x/y)$, so each factor is dominated by $|\log(z_k Q^{k-1})|$, summable by step (3). Dominated convergence gives $G(-R)/\theta_Q(R) \rightarrow 1/[(Q; Q)_\infty \prod_k z_k Q^{k-1}]$ as $R \rightarrow \infty$; evaluating along the subsequence $R = |u|Q^{-N}$ by step (2), the limit equals $1/(q; q)_\infty$, whence $\prod_k z_k Q^{k-1} = (q; q)_\infty / (Q; Q)_\infty = (q; q^2)_\infty$ by Euler's $(q; q)_\infty = (q; q^2)_\infty (q^2; q^2)_\infty$. Finally, in the basis of Proposition 17: along the negative axis (a regular class, $[-1] \notin q^{\mathbb{Z}}$) one has $g^{[\lambda]}(z) \rightarrow 1$ and $f_2(z)/\theta_Q(-z) = O(1/(\theta_Q(R)\theta_Q(R/q))) \rightarrow 0$, so the limit just computed is the entry C_1 ; hence $C_1 = 1/(q; q)_\infty$. The H -statement is identical with comparison function $\theta_Q(-qz)$ and ratio $(1-q)/(q; q)_\infty$. Finally, steps (1)–(2) apply verbatim along $z = uQ^{-N}$ for any $u \in \mathbb{C}^*$ whose orbit avoids the zero class $[1]_Q$ of $\theta_Q(\cdot)$, using the standard theta lower bound $|\theta_Q(-z)| \geq c([u]) \cdot (\text{peak term})$ off that class; along each such orbit $C_1(z)$ is constant (ellipticity) and $f_2/f_1 \rightarrow 0$, $g^{[\lambda]} \rightarrow 1$, so $C_1([u]) = 1/(q; q)_\infty$ on every class $\neq [1]_Q$, and by meromorphy $C_1 \equiv 1/(q; q)_\infty$ — for every regular λ . In particular C_1 is z - and λ -independent. $\square$

Remark 19 (Diophantine consequences at z^*). At $q = q^*$ the identity splits off the candidate rational zero:

$$\prod_{k \geq 2} z_k(q^*) Q^{k-1} = \frac{(q; q^2)_\infty}{z^*} = 0.988299884628746 \dots \quad (\text{verified to 30 digits}).$$

Were β_2 rational, the *modular* value $(q; q^2)_\infty$ (transcendental at algebraic q by Nesterenko's theorem applied to the Euler products) would equal the rational z^* times the tail dressing product $\prod_{k \geq 2} z_k Q^{k-1}$. Thus the entire modular sector of the problem is now explicit — the Stokes-invariant connection constant is $1/(q; q)_\infty$, the zero product is $(q; q^2)_\infty$ — and *all* remaining unknown arithmetic is provably confined to the tail dressing product, equivalently to the q -Stokes cocycle $C_2^{[\lambda]}$: the non-modularity of the β_2 problem *is* its q -Stokes data. A transcendence statement for that tail (an algebraic independence of the higher-zero dressings from the modular values, or a rigidity theorem for the q -Stokes cocycle) would settle β_2 ; no current theorem provides it.

The cocycle itself is now explicit. Writing $g = g^{[\lambda]}$ for the resummed series and h for f_2 's convergent one,

$$C_2^{[\lambda]}(z) = \theta_Q(-z/q) \frac{g(z) G(Qz) + z^{-1} g(Qz) G(z)}{-(z/q) g(z) h(Qz) + z^{-1} g(Qz) h(z)}$$

— an identity: it is the ratio of the Casoratian evaluations $\text{Cas}(f_1, G) = \theta_Q(-z)[g(z)G(Qz) + z^{-1}g(Qz)G(z)]$ and $\text{Cas}(f_1, f_2) = \frac{\theta_Q(-z)}{\theta_Q(-z/q)}[-(z/q)g(z)h(Qz) + z^{-1}g(Qz)h(z)]$, both instances of $\theta_Q(Qw) = w^{-1}\theta_Q(w)$ (and verified numerically to 10^{-61}) — with divisor now fully described: an *explicit* zero lattice $z \in q^{-1}Q\mathbb{Z}$ from the theta factor (the computed zero at $z = 1/q$, exact to 10^{-63}), a direction-dependent zero at $z = \lambda^2$ arising from the exact *resummation resonance* $g^{[\lambda]}(\lambda^2)G(Q\lambda^2) + \lambda^{-2}g^{[\lambda]}(Q\lambda^2)G(\lambda^2) = 0$ (verified to 10^{-62} at $\lambda \in \{\frac{3}{2}, 2, \frac{13}{5}\}$; a proof for general regular λ is open), and poles at the zeros of the h -bracket. Substituting into the pinned zero condition at z^* and cancelling the common theta factor reduces the entire β_2 question to *one* relation between the two modular values $\theta_Q(-z^*)$, $(q; q)_\infty$ and five values of the non-modular pair $(g^{[\lambda]}, h)$ on the orbit of z^* . The remaining input is a transcendence/independence statement for those five values; supplying it is one of three research programs (a non-modular extension of Nesterenko's method via the deformation of this connection, a $\rho < \frac{1}{2}$ -regime irrationality criterion, or adelic rigidity of the vanishing locus); none is currently available.

The first of those programs now has its foundation: the deformation of the module in q closes finitely.

Proposition 20 (the deformation module). *Let $L = q - a\sigma + \sigma^2$ with $a = q + 1 - qz$, $\sigma f(z) = f(Qz)$, and let $\theta = z\partial_z$, $\delta = q\partial_q$ (acting on all q -dependence, including the base $Q = q^2$, so that $\delta\sigma = \sigma\delta + 2\sigma\theta$). Then*

$$L(\theta G) = -qz \sigma G, \quad L(\delta G) = -qG + q(1 + 3z) \sigma G + 4q \theta G - 2a \sigma \theta G$$

(two commutator computations; verified to 51 digits). Consequently the $\mathbb{Q}(q, z)\langle\sigma\rangle$ -module generated by $\{G, \theta G, \delta G\}$ has finite rank ≤ 6 : the two-operator (σ_Q, δ_q) -structure of the β_2 module is finitely generated. Differentiating the pinned zero relation of Proposition 17 along the diagonal $z^* = 2q(1 - q)$ (whose motion contributes θ -terms, which remain in the module) therefore yields a closed quasi-linear system in q for the finitely many non-modular values involved — the precondition for a Nesterenko-type multiplicity method, and the entry point of parametrized q -difference Galois theory (Hardouin–Singer; Di Vizio–Hardouin; Dreyfus–Hardouin–Roques) for this exact module. What that program still requires, and what no current theorem supplies, is the zero-estimate half: a multiplicity bound for the resulting (σ, δ) -ring at (q^*, z^*) . This is the precise, well-posed form in which we leave the β_2 problem.

Corollary 21 (equation of the zero curve). *Implicit differentiation of $G(q, z_1(q)) = 0$ gives*

$$z_1'(q) = - \frac{z_1(\delta G)(q, z_1)}{q(\theta G)(q, z_1)},$$

with $\theta G, \delta G$ in the module of Proposition 20: the zero curve is an integral curve of the rank- ≤ 6 system (formula checked against a finite-difference derivative to 20 digits at $q = 0.3$). The number q^* is the crossing of this curve with the diagonal: $z_1(q^*) = 2q^*(1 - q^*)$ (residual $< 10^{-27}$).

Lemma 22 (irreducibility of the σ -module). *For $0 < q < 1$ the rank-2 σ -module of G is irreducible over $\mathbb{C}(z)$: the Riccati equation*

$$u(z)u(Qz) - a(z)u(z) + q = 0, \quad a = q + 1 - qz,$$

has no solution $u \in \mathbb{C}(z)$.

Proof. Suppose $u = P/R$ with $P, R \in \mathbb{C}[z]$ coprime, of degrees p, r , leading coefficients α, β . Clearing denominators,

$$F(z) := P(Qz)P(z) - a(z)P(z)R(Qz) + qR(z)R(Qz) \equiv 0.$$

At $z = 0$: $F(0) = P(0)^2 - (q+1)P(0)R(0) + qR(0)^2 = 0$; if $R(0) = 0$ then $P(0) = 0$, contradicting coprimality, and symmetrically; so $P(0)R(0) \neq 0$ and $u(0) \in \{1, q\}$. *Top degree*: the three terms have degrees $2p, p+r+1, 2r$. If $p = r$, the top degree $2p+1$ is attained by the middle term alone, with coefficient $q\alpha\beta Q^r \neq 0$; if $|p-r| \geq 2$, the top degree is attained by the first or third term alone, with coefficient $\alpha^2 Q^p$ or $q\beta^2 Q^r$. Both are impossible, so $|p-r| = 1$. *Root pairing*: let $\zeta \neq 0$ be a root of R of multiplicity m , so $P(\zeta) \neq 0$. At $z = \zeta/Q$ the three terms of F have orders $k, k+m+[a(\zeta/Q)=0], m_0+m$, where $k = \text{ord}_{\zeta/Q} P$ and $m_0 = \text{ord}_{\zeta/Q} R$; since $F \equiv 0$ the minimum order cannot be attained by a single term, forcing $k \geq m_0 + m \geq m$. Symmetrically, a root $\xi \neq 0$ of P of multiplicity k forces $\text{ord}_{Q\xi} R \geq k$ (coprimality gives $\text{ord}_{Q\xi} P = 0$, so the first term has order k at ξ , the third $\text{ord}_{Q\xi} R$, and the middle at least $k + \text{ord}_{Q\xi} R$; again no single minimum). The two maps $\zeta \mapsto \zeta/Q, \xi \mapsto Q\xi$ are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities, $p \leq r \leq p$, so $p = r$ — contradicting $|p-r| = 1$. $\square$

Remark 23 (status of the parametrized (δ) -Galois computation). Lemma 22 is the first input to the classification of difference Galois groups of rank-2 q -difference modules: the group is an irreducible subgroup of GL_2 . By the parametrized theory (Hardouin–Singer; Di Vizio–Hardouin; Dreyfus–Hardouin–Roques), an irreducible, non-imprimitive rank-2 module whose equation admits no $\sigma\delta$ -integrability relation has differentially large Galois group, and its solutions are differentially transcendental over $\mathbb{C}(z)$. The imprimitivity exclusion (no rank-1 submodule of the symmetric square) and the integrability certificate for this module are the two remaining verifications; both are finite Riccati-type computations of the same kind as Lemma 22 and are left to a sequel.

Proposition 24 (effective non-rationality of q^*). *The least positive root q^* of S admits no rational representation s/t with $t \leq 10^{191}$. Consequently, if $\beta_2 = u/v$ in lowest terms then $u > 10^{95}$.*

Proof. All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrational`) with the series tail bounded by $(q; q)_{2k} \geq (1-2q)/(1-q) > 0$ for $q < \frac{1}{2}$ and term ratios $< \frac{1}{2}$ beyond the truncation. (i) $S > 0$ on $[0, 0.40]$ (400 certified chunks). (ii) $S' \leq -1$ on $[0.40, 0.46]$ (600 certified chunks; S' evaluated term-by-term with the logarithmic-derivative weights), so S has at most one zero in $[0.40, 0.46]$. (iii) with $a = 0.d_1 \cdots d_{385}$ the 385-digit truncation of q^* and $b = a + 2 \cdot 10^{-385}$, certified evaluation gives $S(a) > 0 > S(b)$; by (i)–(ii), $q^* \in [a, b]$. (iv) the interval $[a, b]$ contains the rational s_0/t_0 (Stern–Brocot minimal fraction) with $t_0 \geq 10^{192}$, verified by exact rational comparison, and certified $S(s_0/t_0) \neq 0$. If $m/n \in [a, b]$ with $n \leq 10^{191}$ and $m/n \neq s_0/t_0$, then $|m/n - s_0/t_0| \geq 1/(nt_0) \geq 10^{-384} > b - a$, impossible; so $m/n = s_0/t_0$, whence $n = t_0 > 10^{191}$, a contradiction. Thus no rational with denominator $\leq 10^{191}$ lies in $[a, b]$ at all, in particular none equals q^* . For $\beta_2 = u/v$ in lowest terms, $q^* = v^2/u^2$ in lowest terms, so $u^2 > 10^{191}$. $\square$

6 The true series U

U inherits the travel poles of Proposition 1; it is transcendental once its numerator is non-zero at infinitely many q_m . The Bousquet–Mélou–type transfer expresses the relevant residue through a Möbius function of a cocycle variable, and non-vanishing reduces to a *gate*

$$b_0 > 0 \quad \text{and} \quad s := \frac{q}{1-q} t_1 < 1, \quad t_1 = \frac{P_{12}}{S_e}, \quad (4)$$

on the off-diagonal cocycle entry P_{12} (see the supplement, § lifting). Both columns of the cocycle satisfy the scalar three-term recursion $y_{n+1} = (1 + q^3 - 2(1 - q)q^{2n+2})y_n - q^3y_{n-1}$; setting $x = q^n$ turns it into the linear q -difference equation

$$Y(qx) + q^3Y(x/q) = (1 + q^3 - 2(1 - q)q^2x^2)Y(x), \quad (5)$$

whose regular solution is, in closed form, a Hahn–Exton q -Bessel function,

$$Y_3(x) = \sum_{k \geq 0} d_k x^{2k+3}, \quad d_k = \frac{(-2)^k (1 - q)^k q^{k^2+3k}}{(q^2; q^2)_k (q^5; q^2)_k} = J_{3/2}^{(3)} \text{ normalised}, \quad (6)$$

and $P_{12} = \frac{2q^3}{1-q^3}Y_3(1)$. The denominator S_e is the block of (1), controlled by Lemma 4. It remains to bound the numerator residual against its confluent limit; this is where the two-Bessel identity enters.

6.1 The amplitude atom, formally verified

As $q \rightarrow 1$, equation (5) degenerates to $\tau Y'' - (2\tau/x)Y' + 2Y = 0$, solved by $x^{3/2}J_{3/2}(Wx)$. The q -corrected connection coefficient across the turning point $X = \nu = \frac{3}{2}$ is governed by the following identity, whose point is that the confluence is *entire* — it passes *through* the turning point, bypassing the Liouville–Green singularity where asymptotic methods stall.

Lemma 25 (two-Bessel confluence; amplitude normalisation). *The regular solution of (5) has the confluent form*

$$Y_3(x) = \gamma_{\text{cl}} x^{3/2} \left[J_{3/2}(X) + \frac{\tau X}{2} J_{5/2}(X) \right] + O(\tau^2), \quad X = wx, \quad \gamma_{\text{cl}} = \frac{3\sqrt{\pi}}{4} \left(\frac{w}{2} \right)^{-3/2},$$

and its single-solution envelope satisfies $A/A_{\text{cl}} = 1 + \tau h(X) + O(\tau^2)$ with the explicit rational

$$h(X) = \frac{1 + 3/(2X^2)}{1 + 1/X^2} \in [1, \tfrac{3}{2}] \quad (X > 0).$$

Consequently $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$ with explicit constant $\frac{3}{2}$.

Proof. From the q -Pochhammer confluence $d_k = \tau^{-k} D_k (1 - k\tau + O(\tau^2))$ with $D_k = (-1)^k / [2^k k! (\frac{5}{2})_k]$, the leading sum is $\gamma_{\text{cl}} x^{3/2} J_{3/2}(X)$ by the Bessel representation of ${}_0F_1$, and the $-k\tau$ correction contributes $\frac{\tau X}{2} \gamma_{\text{cl}} x^{3/2} J_{5/2}(X)$ via ${}_0F_1(\cdot; \frac{7}{2}; \cdot)$. Both series are entire in X . The envelope of $J_{3/2} + \frac{\tau X}{2} J_{5/2}$ follows from $A^2 = M_{3/2}^2 + (\frac{\tau X}{2})^2 M_{5/2}^2 + \tau X (J_{3/2} J_{5/2} + Y_{3/2} Y_{5/2})$ and the closed half-integer cross-term identity

$$J_{3/2}(X) J_{5/2}(X) + Y_{3/2}(X) Y_{5/2}(X) = \frac{4}{\pi X^2} \left(1 + \frac{3}{2X^2} \right),$$

which with $M_{3/2}^2 = \frac{2}{\pi X} (1 + \frac{1}{X^2})$ gives $h(X) = (1 + \frac{3}{2X^2}) / (1 + \frac{1}{X^2})$; this is monotone decreasing from $h(0^+) = \frac{3}{2}$ to $h(\infty) = 1$, so $1 \leq h \leq \frac{3}{2}$. $\square$

Remark 26 (machine verification). The cross-term identity and the bound $1 \leq h(X) \leq \frac{3}{2}$ have been formalised and checked in Lean 4 against Mathlib (file `AtomN.lean`); the proof terms depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The identity is proved by a single `linear_combination` off $\sin^2 + \cos^2 = 1$, and the bound by `nlinarith`.

Theorem 27. *Suppose the numerator amplitude estimate $(\star)$ holds: at every travel pole q_m the amplitude $Y_3(1/q_m)$ agrees with its confluent leading part to relative order $O(\tau^2)$ — equivalently $|P_{12}(q_m)| = O(\tau^{3/2})$. Then the true growth series U is transcendental over $\mathbb{Q}(x)$. Every other analytic input coincides with Theorem 8's: the reduction to the numerator gate (4) is exact and machine-checked, and the denominator half of the gate is Lemma 4, shared with V . The estimate $(\star)$ is derived to leading order in Remark 7 — phase-matching the amplitude's zero against the travel-pole equation gives $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$, whence $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the gate threshold $1/\sqrt{2}$ — and it holds at the first 70 poles; its residual $O(\tau)$ term (the amplitude drift from turning point to pole) is established numerically but not proved, and is the sole gap (Remark 28).*

Proof. At a travel pole write $t_1 = (E/E_S)(1+R/E)/(1+R_S/E_S)$ where $E = \frac{1}{2}(w-W)^2 \sin w \sin(w-W)$ is the elementary confluent leading part of P_{12} , $E_S = \sin w \sin(w-W)$ that of S_e , and $R = P_{12} - E$, $R_S = S_e - E_S$ the residuals. The leading ratio $E/E_S = \frac{1}{2}(w-W)^2 \rightarrow \frac{\tau}{4}$ is exactly elementary, so $s \rightarrow \frac{1}{4} < 1$. *Denominator.* $R_S = \cos w \cos(w-W) - T_2$, and by Lemma 4 $|T_2| = O(\sqrt{\tau})$ with the small constant $\sqrt{2}/36$, so $S_e = E_S + R_S$ stays bounded away from zero (the same margin as in Theorem 8) and R_S/E_S is bounded — exactly as for V . *Numerator.* The remaining bound $R/E = O(\tau)$ is equivalent to $|P_{12}| = O(\tau^{3/2})$ at the pole; since $P_{12} = \frac{2q^3}{3(1-q^3)}Y_3(1/q) - \frac{2}{3}S_e$ is an $O(\sqrt{\tau})$ cancellation, it demands $Y_3(1/q)$ at $X = w$ to relative order $O(\tau^2)$. This bound holds at every computed pole ($|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.1768$, a $4\times$ margin below $1/\sqrt{2}$, over $m = 4, \dots, 70$), but is established there *numerically*; it is exactly the hypothesis $(\star)$. Under $(\star)$, $t_1/\tau = \frac{1}{4} + O(\tau)$ and $s < 1$ with positive margin at every pole, so $\mathcal{N}_U(q_m) \neq 0$, and U inherits infinitely many uncanceled poles accumulating at $q = 1$, hence is transcendental. $\square$

Remark 28 (the remaining gap). Lemma 25 discharges the amplitude *through the turning point* $X = \frac{3}{2}$ — the obstruction to any Liouville–Green treatment, and the one piece formerly thought to settle the U -side. It does *not*, however, supply the amplitude at the *pole* $X = w$ to the $O(\tau^2)$ -relative accuracy the numerator gate requires. There the two-Bessel form carries a relative defect whose $O(1)$ part is exactly $\frac{1}{36}$ — i.e. the classical turning-point-to-pole connection factor $\frac{36}{35}$, which is itself derivable and, once applied, matches $Y_3(1/q)$ to $O(\tau)$ -relative — and whose *residual* $O(\tau)$ part (the amplitude drift from the turning point to the pole) remains unproven. Because $P_{12} = \frac{2q^3}{3(1-q^3)}Y_3(1/q) - \frac{2}{3}S_e$ is an $O(\sqrt{\tau})$ cancellation, that residual is amplified to the size of P_{12} itself; only $Y_3(1/q)$ to $O(\tau^2)$ -relative — a uniform pole-amplitude resummation of the *same analytic class* as Lemma 4 — closes it. Crucially, this residual is of *exactly the denominator's kind*, not a harder one. The two off-diagonal cocycle entries obey the Abel relation $x_N Y_N - X_N y_N = -1$ (the discrete q -Wronskian, conserved because $\det M_n \equiv 1$), whence

$$P_{12} = \frac{X_N S_e - 1}{x_N} \quad \text{exactly,}$$

x_N, X_N being the leading components of the same three-term recursion. Hence P_{12} is fixed by the *same* family of Hahn–Exton blocks that controls S_e : one analytic class — that of Lemma 4 — not two. In this precise sense U sits on V 's footing. What U additionally requires is the pole asymptotics of x_N, X_N — again Hahn–Exton, again governed by the exact q -Wronskian of Olde Daalhuis [6], $W_q(J_\nu, Y_\nu) = q^{\nu(\nu-1)}(1-q^2)/\Gamma_{q^2}(\frac{1}{2})^2$ for the numerically satisfactory pair, whose $q \rightarrow 1$ expansion is explicit via Moak's q -Stirling formula (e.g. $\Gamma_{q^2}(\frac{1}{2}) = \sqrt{\pi}(1 - \frac{3}{8}\tau + O(\tau^2))$). The block-asymptotic assembly is carried out *elementarily* in Remark 7: the numerator amplitude has a derived closed-form expansion with exact rational constants, and the phase-matching of the

travel-pole equation against the amplitude's zero — both collapsing to the same sinh-product family — derives the gate value $1/(4\sqrt{2})$. U 's analytic inputs thereby coincide with V 's up to the $O(\tau)$ residual of $(\star)$. (The orthogonal arithmetic route of §7 would settle U without any of this analysis, but is itself open.)

7 An orthogonal arithmetic route

For each prime p , $U \bmod p \in \mathbb{F}_p[[q]]$ is algebraic over $\mathbb{F}_p(q)$ iff its coefficient sequence is p -automatic (Christol [5]), iff its p -kernel is finite. We find the p -kernel to be *maximally* non-automatic: through the accessible depth its size tracks the full p -ary tree with *no* deficit. (An apparent deficit of 1 at level 3 — 1, 4, 13, 39 vs. the full 1, 4, 13, 40 at $p = 3$ — is a *comparison-length artifact*: a level- k decimation subsequence retains only $\sim N/p^k$ terms, so at a short prefix two distinct decimations can coincide; the deficit disappears once the comparison length reaches 9.) Two instances are machine-verified in Lean 4. For the true series itself, the 3-kernel of $(u_n \bmod 3)$ through level 3 is the full ternary tree, deficit 0: all $1 + 3 + 9 + 27 = 40$ decimation words of length ≤ 3 are pairwise distinct — at comparison lengths 9 and 10 alike — on the 271-term prefix $n \leq 270$ (`UKernel.lean`, from the validated `tools/u_modp_rust` output, whose u_n are self-checked against the 43 known terms and byte-identical to the earlier validated $n \leq 180$ run on the overlap). For the sibling block Σ_1 , already transcendental over $\mathbb{Q}(q)$ by Theorem 2, the same level-3 full tree is certified in `SigmaKernel.lean` (built from the A/C recursion over $\mathbb{F}_3$), with the short-prefix deficit-1 confirmed spurious. The Berlekamp–Massey linear complexity of $(u_n \bmod 3)$ is $\approx N/2$. This is the strongest finite-data evidence for non-automaticity, and hence for transcendence over $\mathbb{Q}(x)$, independent of all of §§4–6. A proof for U would require a non-automaticity criterion valid for *dense*-support series — the coefficients carry no lacunary gaps — which the standard gap/Mahler methods do not provide.

8 Conclusion

The arithmetic of A396406 separates cleanly. The catalytic building blocks and the growth rate are unconditional (Pólya–Carlson; Corollary 9). Both the relaxed series V and the true series U rest on a single simple-saddle steepest-descent estimate (Lemma 4), which we now prove *self-contained* — by Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the explicit bounded-variation bound of Lemma 3, the two saddles $z_{\pm}^* = \pm iW/2$ supplying the global cancellation that no partial-summation method can — rather than by citing a general theorem. With that estimate in hand, **V is transcendental over $\mathbb{Q}(x)$ unconditionally**. The true series U stands on the same footing up to a single residual: through the exact identity $P_{12} = \frac{2q^3}{1-q^3} \sum_k d_k$, its one further input — the numerator amplitude at the pole, estimate $(\star)$ — is derived in elementary closed form to leading order, and the phase-matching of the travel-pole equation against the amplitude's zero gives $\delta w = \frac{\sqrt{2}}{8} \tau^{3/2}$, hence $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) < 1/\sqrt{2}$: the gate holds with a fourfold margin, established to leading order and at the first 70 poles. **Conditional on the $O(\tau)$ residual of $(\star)$ — verified numerically but not proved — U is transcendental over $\mathbb{Q}(x)$** with the same analytic standing as V : every coefficient derived elementarily, remainders at the standard simple-saddle level of Lemma 4. The turning-point amplitude atom on the U -side is the elementary rational bound of Lemma 25, formally verified in Lean 4. The independent mod- p route of §7 would settle U by a completely different mechanism, if a dense-support non-automaticity criterion can be found.

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